



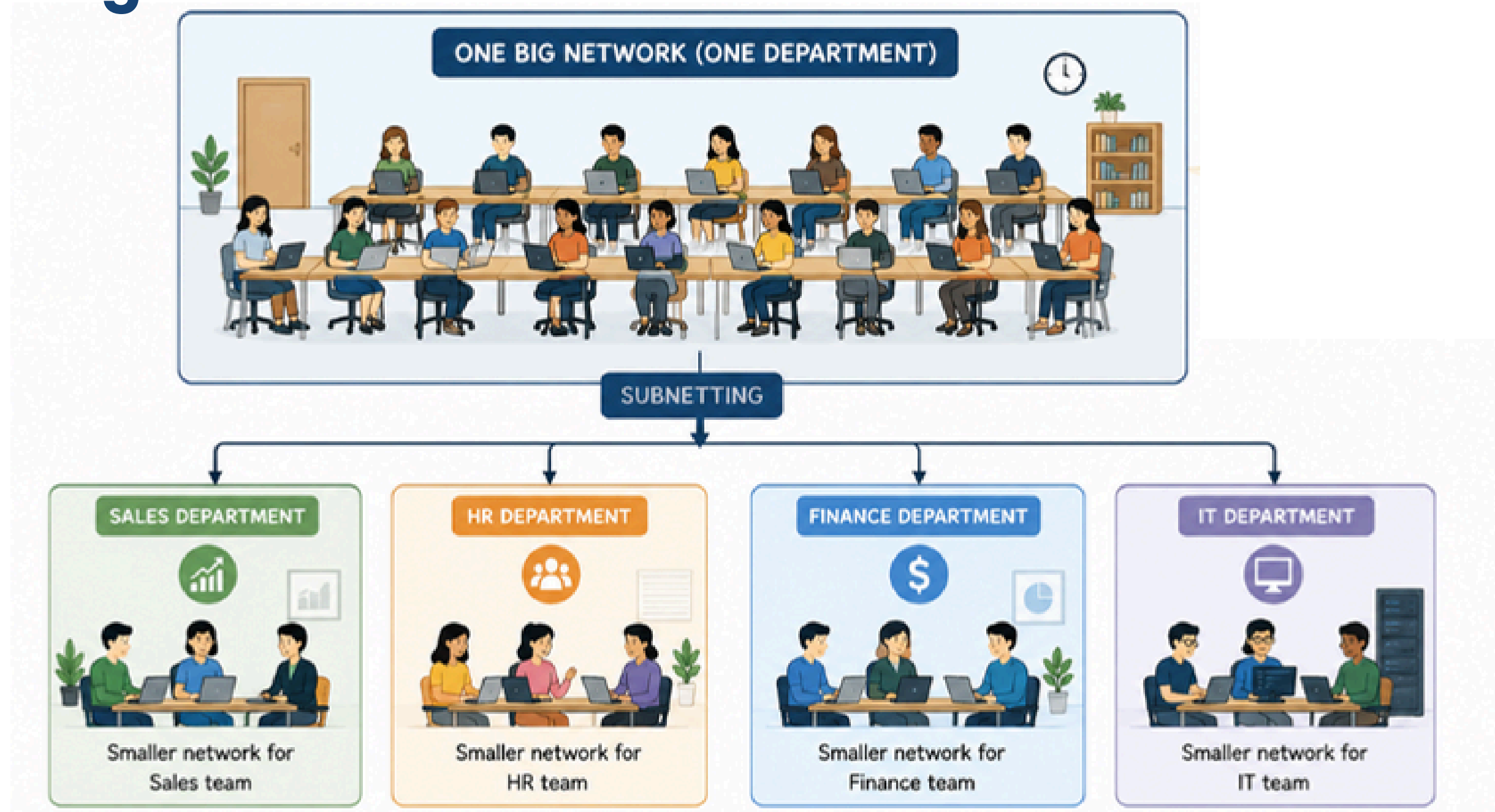
# **NAT, PAT & DHCP — and How VPC Route Tables Wire It Together**

by Vikas Kumar

CSAI Computer Networks

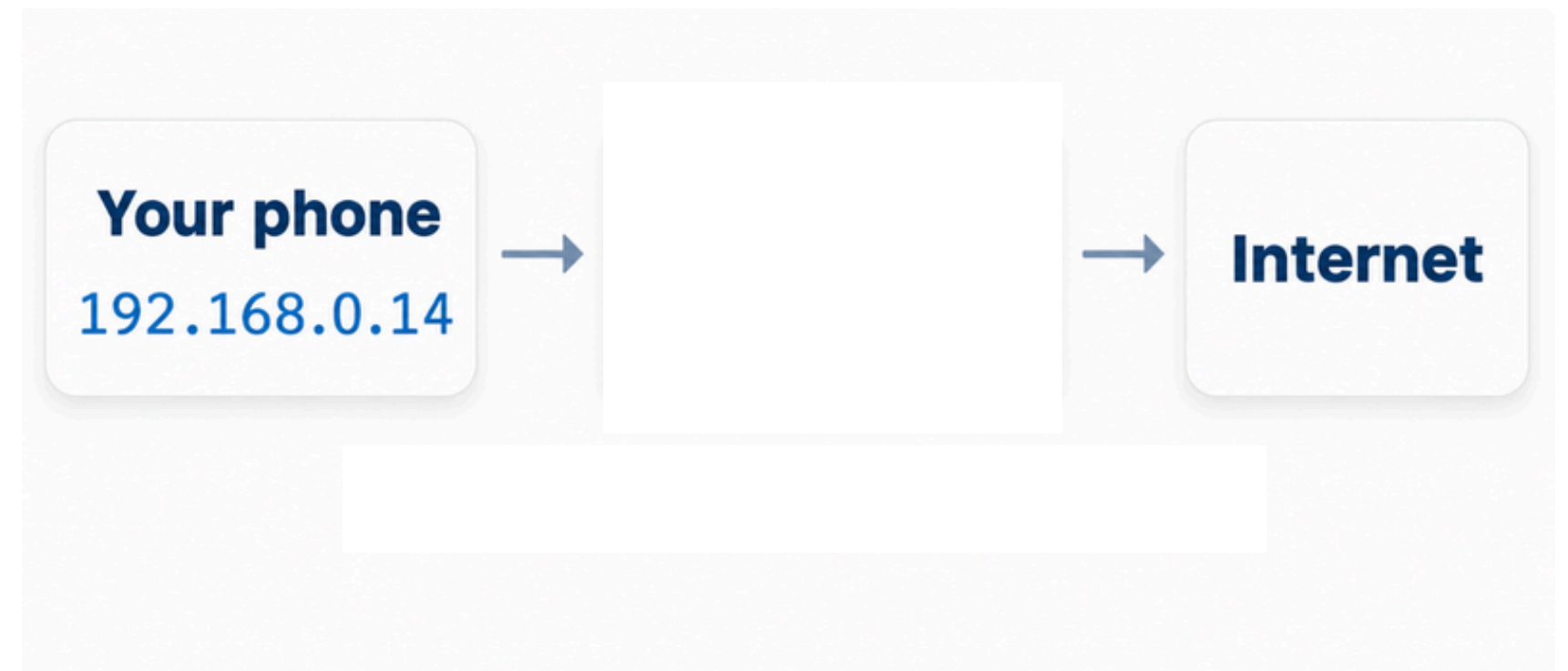
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# Recap Subnetting



# How Does Your Phone Browse on a Private IP?

- Private subnets can't be reached from the Internet.
- Yet your phone on home WiFi has a private IP...
- ...and still loads every website just fine.

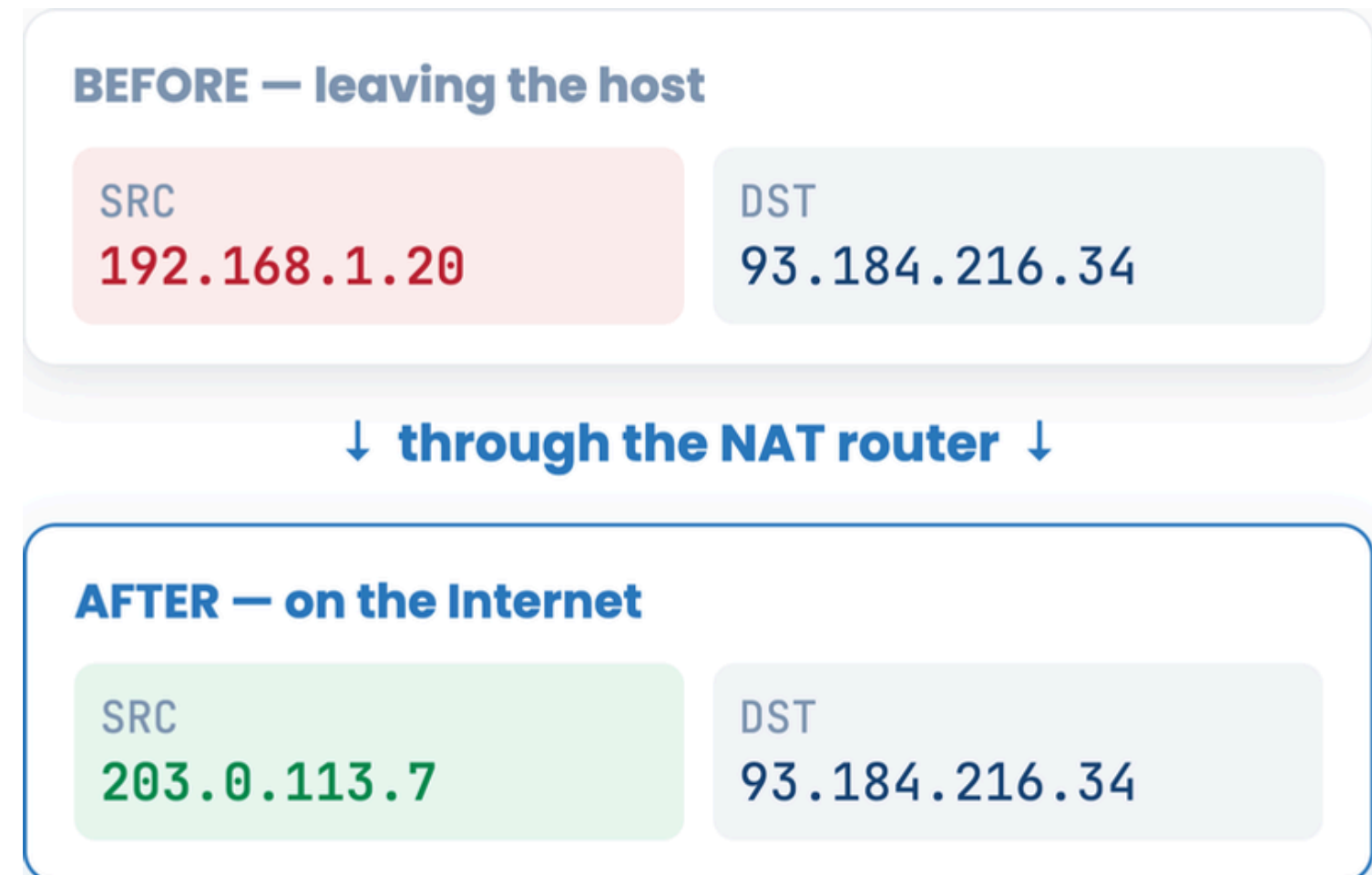


# NAT

*Rewriting addresses at the border.*

# What Does NAT Actually Do?

- On the way out, the router swaps the private source IP for a public one.
- On the way back, it reverses the swap, restoring the private IP.



*The reply only finds its way home because the router remembers the swap.*

# Ten Devices, One Public IP — All at Once?



| Inside (private) | Outside (public) | Destination       |
|------------------|------------------|-------------------|
| 192.168.1.20     | 203.0.113.7      | 93.184.216.34:443 |
| 192.168.1.21     | 203.0.113.7      | 93.184.216.34:443 |

*Everyone streaming, scrolling, gaming — through a single address. How?*

# PAT

*Rewriting addresses at the border.*

# PAT– Port Address Translation

- Add the port number to the translation — not just the IP.
- One public IP can now serve many hosts at the same time.
- This is exactly what your home router does every second.



***Same public IP, a different port each — that is the whole trick.***

# What's Inside the Translation Table?

| Inside (private)   | Outside (public)  | Destination       |
|--------------------|-------------------|-------------------|
| 192.168.1.20:51000 | 203.0.113.7:40001 | 93.184.216.34:443 |
| 192.168.1.21:51000 | 203.0.113.7:40002 | 93.184.216.34:443 |
| 192.168.1.35:49876 | 203.0.113.7:40003 | 142.250.72.14:443 |

*Two hosts reused port 51000 — the outside port is the key that routes each reply home.*

# What Do We Gain, What Do We Break?

## We gain

- Conserves scarce public IPv4 addresses.
- Hides internal hosts — a security side-effect.

## We break

- End-to-end reachability is gone.
- Inbound now needs explicit setup.

**Two laptops behind one home IP both open the same website. What field lets the NAT box tell their replies apart?**

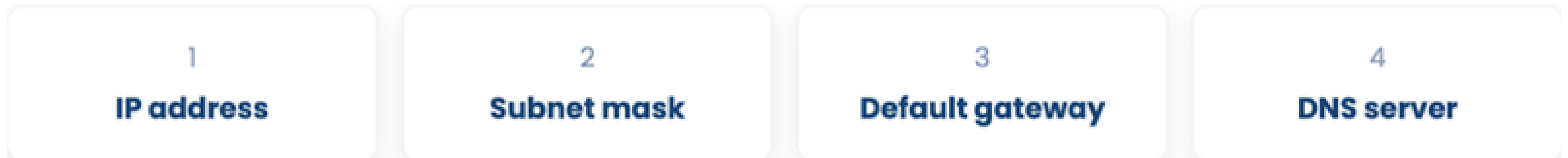
**The port number — that's PAT.**

# DHCP

*Addresses, handed out automatically.*

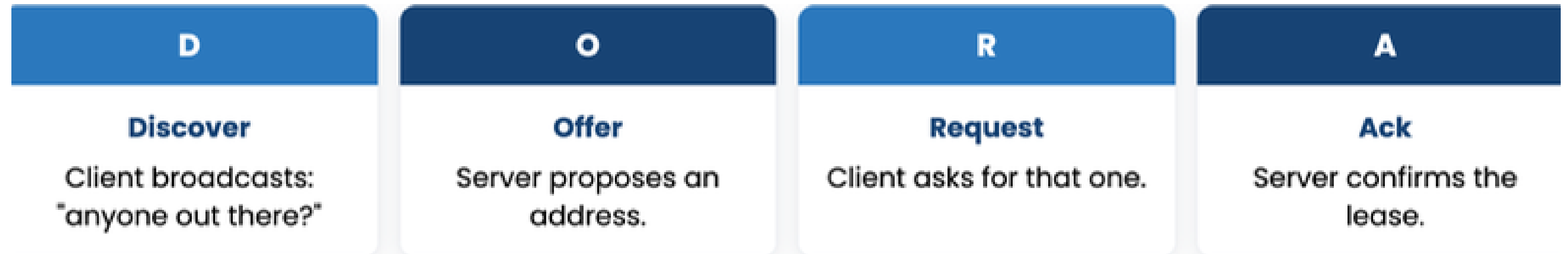
# Who Configures Every New Device?

Nobody types IP settings by hand. One DHCP lease delivers four things at once:



*A lease expires — so addresses get renewed and recycled over time.*

# Meet DORA: The Four-Step Handshake



*Discover and Request are broadcasts — the whole subnet hears them.*

# What Does DORA Look Like on the Wire?

- Capture on any subnet and you'll see the four frames.
- Discover and Request go to 255.255.255.255.
- The address comes from a Lecture 14 subnet range.

| dhcp-capture.pcapng |             |                 |       |          |
|---------------------|-------------|-----------------|-------|----------|
| No.                 | Source      | Destination     | Proto | Info     |
| 1                   | 0.0.0.0     | 255.255.255.255 | DHCP  | Discover |
| 2                   | 192.168.1.1 | 255.255.255.255 | DHCP  | Offer    |
| 3                   | 0.0.0.0     | 255.255.255.255 | DHCP  | Request  |
| 4                   | 192.168.1.1 | 255.255.255.255 | DHCP  | Ack      |

*The clearest broadcast handshake you'll capture all term.*

# Routing Tables

*The decision at every hop.*

# What Is a Route Table, Really?

- Just a list: destination prefix → next hop.
- The router checks it for every single packet.
- The most specific match wins — longest-prefix match.

| Destination | Next hop |
|-------------|----------|
| 10.0.0.0/16 | local    |
| 10.0.1.0/24 | router-B |
| 0.0.0.0/0   | gateway  |

***No magic — a lookup table the router reads by specificity, top match wins.***

# Where Does "Everything Else" Go?

- 0.0.0.0/0 is the default route — the catch-all.
- When prefixes overlap, the longest one wins.
- So where does a packet to 10.0.1.55 go?

Packet to 10.0.1.55 — which route?

|             |                |
|-------------|----------------|
| 0.0.0.0/0   | /0 · too broad |
| 10.0.0.0/16 | /16 · matches  |
| 10.0.1.0/24 | ✓ winner (/24) |

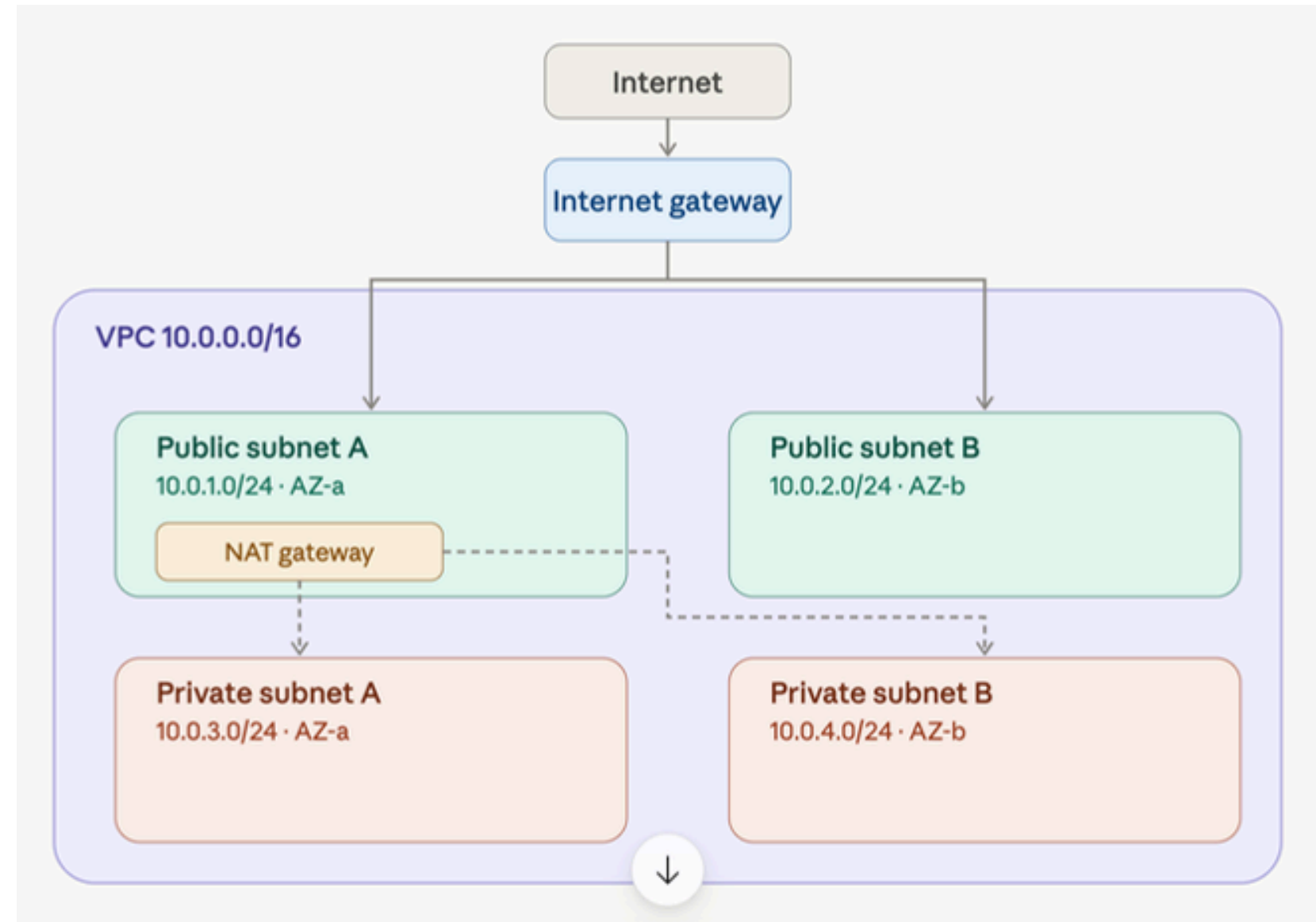
*Specificity beats generality — the longest prefix wins, every time.*

# The AWS Picture

*IGW, NAT Gateway & VPC route tables.*

# How Does a VPC Touch the Internet (IGW)?

- An IGW attaches your VPC to the Internet.
- A public subnet routes 0.0.0.0/0 → IGW.
- Its instances can be reached both ways.



***"Public subnet" simply means: has a route to the IGW.***

# How Does a VPC Touch the Internet (IGW)?



***"Public subnet" simply means: has a route to the IGW.***

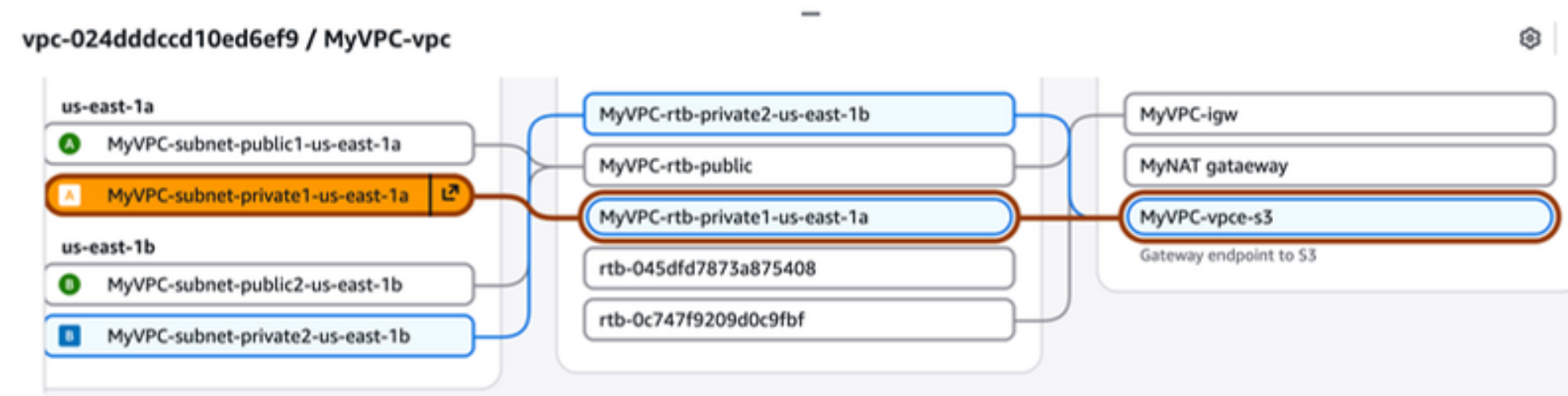
# How Does a VPC Touch the Internet (IGW)?

Route table: [rtb-0501e38d89e6a39fa](#) / MyVPC-rtb-public

| Routes (2)                                     |                                       |
|------------------------------------------------|---------------------------------------|
| <div><div></div><div>Filter routes</div></div> |                                       |
| Destination                                    | Target                                |
| 10.0.0.0/16                                    | local                                 |
| 0.0.0.0/0                                      | <a href="#">igw-0e4226802ca4594de</a> |

# Outbound Only — The NAT Gateway

- The NAT Gateway lives in a public subnet.
- A private subnet routes 0.0.0.0/0 → NAT GW.
- Instances get updates out — but stay unreachable in.



*Updates flow out; attackers can't flow in.*

# Outbound Only — The NAT Gateway

|                                     |                                  |                                           |           |                                                |     |
|-------------------------------------|----------------------------------|-------------------------------------------|-----------|------------------------------------------------|-----|
| <input type="checkbox"/>            | -                                | <a href="#">subnet-0622c220c27001222e</a> | Available | <a href="#">vpc-0ee9879991d5196a6</a>          | Off |
| <input checked="" type="checkbox"/> | MyVPC-subnet-private1-us-east-1a | <a href="#">subnet-0d224d45b2001770e</a>  | Available | <a href="#">vpc-024dddccd10ed6ef9   MyV...</a> | Off |
| <input type="checkbox"/>            | MyVPC-subnet-private2-us-east-1b | <a href="#">subnet-0392ace67a3250fc7</a>  | Available | <a href="#">vpc-024dddccd10ed6ef9   MyV...</a> | Off |
| <input type="checkbox"/>            | -                                | <a href="#">subnet-000e1357ff9698d3e</a>  | Available | <a href="#">vpc-0ee9879991d5196a6</a>          | Off |
| <input type="checkbox"/>            | -                                | <a href="#">subnet-017a14e7b08417ca2</a>  | Available | <a href="#">vpc-0ee9879991d5196a6</a>          | Off |

subnet-0d224d45b2001770e / MyVPC-subnet-private1-us-east-1a

Route table: [rtb-0715f7aa8f1d9415e](#) / MyVPC-rtb-private1-us-east-1a

[Edit route table associa](#)

## Routes (2)

< 1 >

Destination



Target

10.0.0.0/16

local

[pl-63a5400a](#)

[vpce-06cd2371b631a5c0a](#)

*Updates flow out; attackers can't flow in.*

# Public vs. Private Route Table — Spot the Difference

## Public subnet RT

|             |          |
|-------------|----------|
| 10.0.0.0/16 | local    |
| 0.0.0.0/0   | igw-0a1b |

## Private subnet RT

|             |          |
|-------------|----------|
| 10.0.0.0/16 | local    |
| 0.0.0.0/0   | nat-9f7c |

***Same VPC. One word in one row changes who reaches the Internet — and who reaches you.***

**A private-subnet server needs OS updates but must never accept inbound connections. Which gateway, and what route?**

NAT Gateway · 0.0.0.0/0 → NAT GW

# Activity: Be the NAT Box

*Five minutes. One translation table. Everyone plays.*

# Be the NAT Box — Then Break It

- 1 Cast the room: one NAT router at the board, four inside hosts with private IPs, instructor = the Internet.
- 2 Each host sends a request card; the NAT student rewrites source → one public IP + a unique port, and logs it.
- 3 Replies come back to the public IP + port — route each one home using only the table.
- 4 The twist: reuse a port on purpose → collision → feel exactly why every mapping must be unique.

***Same logic you just performed runs as the AWS NAT Gateway in the console.***

# Key Takeaways

- NAT/PAT lets many private hosts share one public IP by rewriting address + port — the reason your whole house browses through one ISP address.
- DHCP (DORA) auto-leases address + mask + gateway + DNS from your subnet's range.
- In AWS the route table is the switch: 0.0.0.0/0 → IGW makes a subnet public; 0.0.0.0/0 → NAT GW gives a private subnet outbound-only Internet.

# Next Up: ICMP, then IPv6

- We've leaned on ping & traceroute all term — they ride on ICMP.
- Next: ICMP properly, and the diagnostics it powers.
- Then IPv6 — the long-term answer to the exhaustion that made NAT necessary.

**Please fill the feedback form.**

**Thank You**